The empirical recurrence for OEIS A253478: recovering a conjecture that is not written down

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Abstract

OEIS A253478 records "Empirical recurrence of order 88", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 114$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 88, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 88 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A253478 is "Number of (n+1)X(2+1) 0..2 arrays with every 2X2 subblock ne-sw antidiagonal difference unequal to its neighbors horizontally and nw+se diagonal sum unequal to its neighbors vertically.". It has offset 1 and begins

$$558, 6804, 85280, 1066300, 13347930, 166986430, 2089637292, 26146542588, \ \dots$$

The entry states:

> Empirical recurrence of order 88 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 14:03:18, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 586$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 114$ of the 586, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 114.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1172$ exact terms of the model returns order 88 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{88} c_i\, a(n-i),$$

c_{1}	3	c_{23}	191595568292	c_{45}	3071753332165228	c_{67}	6304821985462048
c_{2}	110	c_{24}	−351766961468	c_{46}	−1996264711322523	c_{68}	−3050346785266524
c_{3}	242	c_{25}	−195854600597	c_{47}	−5978895963608458	c_{69}	−851386388109060
c_{4}	−967	c_{26}	−982531182910	c_{48}	3456758898243515	c_{70}	2094548312264168
c_{5}	−4962	c_{27}	−1605952445792	c_{49}	7517970566642611	c_{71}	−2791015804165192
c_{6}	−36887	c_{28}	7538481051182	c_{50}	−17022516510779176	c_{72}	1847845145082228
c_{7}	−70132	c_{29}	8552848953171	c_{51}	15190288609726494	c_{73}	−1301018622926192
c_{8}	426277	c_{30}	−6828333113116	c_{52}	7874401293288032	c_{74}	698708347800624
c_{9}	1013386	c_{31}	−12722180045000	c_{53}	−20974299978357991	c_{75}	−140871895899264
c_{10}	1395849	c_{32}	−42810788909448	c_{54}	17346159659592811	c_{76}	29832403616384
c_{11}	5640281	c_{33}	−35075266587511	c_{55}	−12326163763086535	c_{77}	168607574322688
c_{12}	−29936926	c_{34}	94746480273624	c_{56}	−1521066758209064	c_{78}	−171389227032832
c_{13}	−91118256	c_{35}	163233342821720	c_{57}	7576094681984354	c_{79}	157906080546816
c_{14}	26993420	c_{36}	40028779957887	c_{58}	−3366933751623679	c_{80}	−107244297793536
c_{15}	−14229570	c_{37}	−79033529762991	c_{59}	162514254701062	c_{81}	49196713271296
c_{16}	917377417	c_{38}	−392465223966635	c_{60}	1518420089789147	c_{82}	−23839744311296
c_{17}	2880957476	c_{39}	−111180624340288	c_{61}	3105675269656268	c_{83}	7437776977920
c_{18}	−4054475589	c_{40}	−15137085688351	c_{62}	−2950645356799468	c_{84}	−2494595399680
c_{19}	−6193181699	c_{41}	−474621903577420	c_{63}	−663436330887759	c_{85}	457048588288
c_{20}	−6200523589	c_{42}	2124206007829863	c_{64}	1898362473159842	c_{86}	−53237252096
c_{21}	−35617320100	c_{43}	−543026006410527	c_{65}	1470149629707508	c_{87}	−10548674560
c_{22}	121589454290	c_{44}	−1046215836176064	c_{66}	−5740832813172777	c_{88}	−570425344

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 88, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $88 + 4$ terms removed returns order 88 as well, so $\deg q_{\min} = 88 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $88+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 88 that this sequence satisfies — and that family turns out to have exactly one member.

References

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